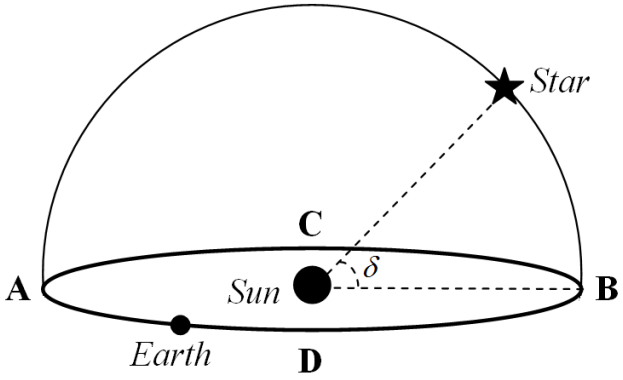


φ . δ
D , $\pi/2$ B , $\pi-\delta$ C .
$\pi/2$ A .

$$a_1 = \frac{V}{c} \tag{16}$$

$$a_2 = \frac{V}{c}\sin\delta\,. \tag{17}$$

$$\delta = \arcsin\left(\frac{a_2}{a_1}\right) = 64.2^\circ\,. \tag{18}$$

2.4 **[2,2점]** (12)
$\varphi = 0$

$$\left(\frac{\Delta\omega}{\omega}\right)_D = 1-\sqrt{\frac{1-v_X/c}{1+v_X/c}} \approx 9.95\times10^{-3}. \tag{19}$$

.

.

$$m_{ph} = \frac{\hbar\omega}{c^2}, \tag{20}$$

r

$$F = G\frac{m_{ph}M}{r^2}\,. \tag{21}$$

$$\hbar d\omega = -F dr\,. \tag{22}$$

$$\frac{d\omega}{\omega} = -\frac{GM}{c^2}\frac{dr}{r^2}\,. \tag{23}$$

(B4.5) R ∞

$$\ln\left(\frac{\omega}{\omega_0}\right) = -\frac{GM}{c^2R}, \tag{24}$$

ω_0 ω
.